* Understanding of Problem statement
* Data Ingestion
* Data Preparation | Preprocessing

- Missing value treatment
- Class Imbalance
- Outlier treatment
- Data Encoding
- Feature Engineering / Feature Extraction.

Feature Extraction. / F E

- Creating new features
- Modifying the exist feature
- Selecting the right feature.

* Feature Extraction → process of selecting and extracting the relevant features from the raw data.

1000 features

f_1 (#rooms)	f_2 (Area)	...	f_{1000} (Sex)	y (Price)
			M	
			F	

* All the features can not be relevant for machine learning models.

↓

How it can be a problem?

* Curse of dimensionality

↳ With increase in no of features :-

① Model training becomes computationally expensive.

② Model interpretation becomes complex.

How to solve?

① Creating new features

Ex1

distance	time	speed = Distance / Time
50	2	50/2
100	2.5	100/2.5
150	3.5	—
110	—	—
—	—	—
—	—	—
—	—	—

Ex2

temp	3 months moving avg.
21°	Nan
22°	Nan
23°	(21+22+23)/3
18	(22+23+18)/3
20°	→
25°	→
.	

No. of rooms = effective area / Area of room

Ex-3.

Area of room	No. of rooms	Effective area
100	2	200
200	3	600

② Modifying the existing feature

Ex-1

Age	Modified Age
15yrs	15
20yrs	20
25yrs	25
⋮	

ⓐ Changing the datatype.

Ex-2.

install	install modified
APP-1 15f	15
APP-2 20f	20
APP3 25f	25

Ex3

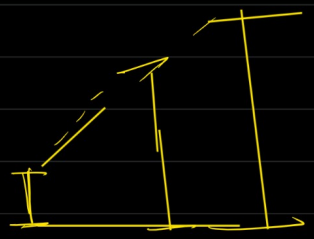
Date	Day	Month	Year
28-10-2024	28	10	2024
26-10-2024	26	10	2024
	⋮	2	2022
	⋮	3	⋮
	⋮	1	⋮

⑤ feature Scaling (optional)

Thousands → ones → tens

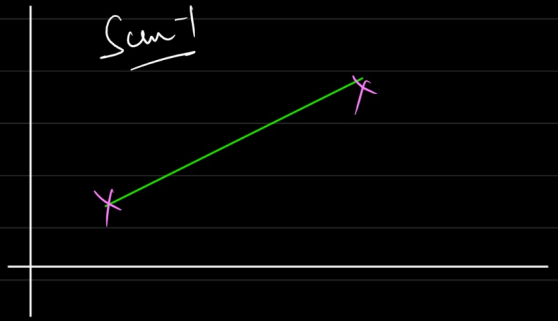
Area of house	# of rooms	Parking area	Y (price of house)
1800	2	100	60
2500	3	50	65
3000	4	25	100
6000	1	30	⋮
⋮	5	⋮	⋮
⋮	⋮	⋮	⋮

* feature Scaling tries to bring all the features on the same Scale.



Why ?

Scn-1



Scn-2



* Many of the Algorithms are distance based, that why if high magnitude it comes computationally expensive.

↳ optimisation becomes faster.

* Interpretation becomes easier.

Types of scaling

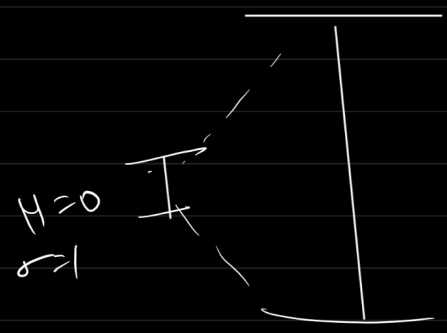
① Standardisation (ML algorithms)

$$Z_{score} = \frac{x_i - \bar{x}}{\sigma} \Rightarrow SND = \mu = 0, \sigma = 1$$

Age	Agscaled
25	$\frac{25-24.6}{1.03}$
26	$\frac{26-24.6}{1.03}$
23	—
24	—
25	—

$\bar{x} = 24.6$

$\sigma = 1.03$



② Normalisation [min-maxscaler] $\Rightarrow [0,1]$ → DL Algorithms → PCA.

Age	
25	$\frac{25-23}{26-23} = 2/3$
26	$\frac{26-23}{26-23} = 1$
23	—
24	—
25	—

min = 23

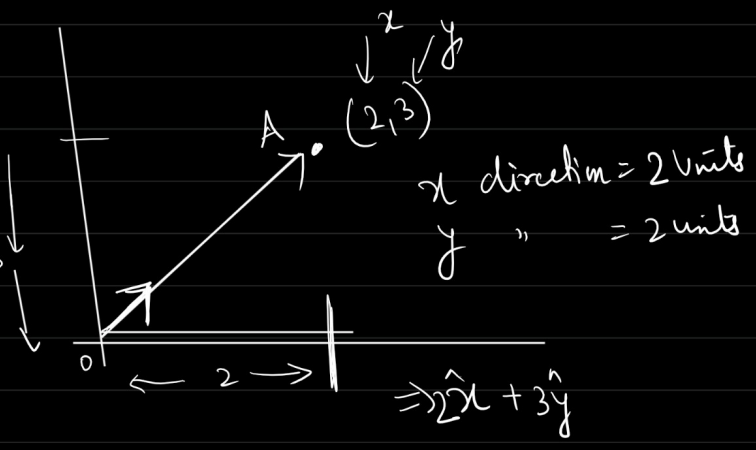
max = 26

$$x_{scaled} = \frac{x_i - x_{min}}{x_{max} - x_{min}}$$

$\downarrow$

$[0,1]$

③ Unit Vector



Pythagoras theorem

$$OA^2 = OB^2 + AB^2$$

$$OA = \sqrt{2^2 + 3^2}$$

$$\|\vec{OA}\| = \sqrt{13}$$

$$\hat{u} = \left(\frac{2}{\|\vec{OA}\|}, \frac{3}{\|\vec{OA}\|} \right)$$

Unit vector.

$$\sqrt{\left(\frac{2}{\sqrt{13}}\right)^2 + \left(\frac{3}{\sqrt{13}}\right)^2}$$

$$= \sqrt{\frac{4}{13} + \frac{9}{13}} = \sqrt{1} = 1$$

③ Selecting the right feature (feature selection)

1000 → Top 10 features

↓

training of model.

① Filter method

↳ Calculate Correlation with target Variable

if this is more

Corr (Area of house, Price of house)

Corr (Area of Parking, Price of house)

x_1 Area of House	x_2 Area of Parking	y Price of house

② Embedded method

③ Wrapper method

* PCA.

df['total_bill']

Standardisation

• fit → mu, std (Calculate mu, std)

• transform → $\frac{x_i - mu}{std}$ (apply the calculation on dataset using mu, std)

(model training)

train

rows	data		
1	—	—	—
2	—	—	—
3	—	—	—
4	—	—	—
5	—	—	—

test (testing the model) ← representative of Unseen data

↓

We don't know this data.

Scaling (Standardisation)

- mu, sigma (fit) → train
- apply $\frac{x_i - mu}{\sigma}$ (transform) → train, test

fit & Transform → Train

Transform → Test

* Scaling is optional & should be done after train-test split.